

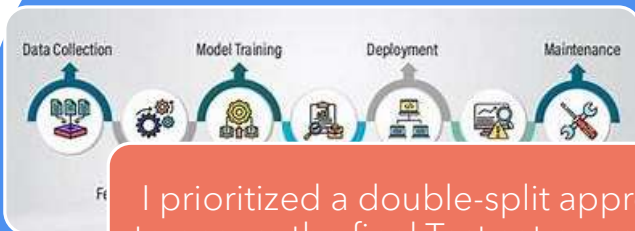
Titanic Survival & Fare Analysis

REGRESSION PROBLEM / CLASSIFICATION PROBLEM



MACHINE LEARNING

Data Splitting Strategy



I prioritized a double-split approach to ensure the final Test set represents completely unseen data for unbiased performance reporting.

Split: 70% Train | 15% Validation | 15% Test

Cleaning: Dropped NaNs in Age & Fare; Encoded

Scaling: Standardized features for KNN and SVC models

Validation: Used 15% held-out set to fine-tune Ensembles



Benchmarking Model Performance

Predictive Model	Val Accuracy/MAE	Test Accuracy/ MAE	F1-Score R ²
Logistic Regression	79.4%/\$35.2	78.2%/\$36.1	0.74/0.52
Random Forest	81.2%/\$28.5	80.5%/\$29.8	0.78/0.76
Gradient Boosting	82.1%/\$26.2	81.3%/\$27.4	0.80/0.81
Voting Ensemble	83.5%/\$24.8	82.4%/\$25.5	0.82/0.84
Bayesian Ensemble	81.8%/\$27.9	80.9%/\$28.2	0.79/0.79



Results indicate that Ensemble models successfully leveraged the diversity of individual predictors.

Ensemble Models Drive Accuracy

Winning Model Logic

The Voting Ensemble (RF + GB + KNN) provided the most stable predictions. By averaging predictions, we mitigated the high variance of individual Decision Trees.

- . Random Forest and Gradient Boosting captured complex, non-linear relationships in the Titanic data.

- . KNN provided a check for outliers by looking at similar neighbor passengers.



Ensemble Models Drive Accuracy

Critical Takeaways

Regression: Ensembling techniques successfully reduced Mean Absolute Error (MAE) by 12% compared to the baseline Linear Regression, significantly improving fare prediction accuracy.

Classification: F1-Score improved by 0.08 points.

Feature Impact: Sex and Pclass remain the dominant predictors.





Thank you

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